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# Thermodynamics and Statistical Physics

Spring 2026

## Homework 11

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Show all essential steps in your derivations.

### Mean-field theory from the variational principle

Consider the Ising model

$$H(\sigma) = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i, \quad \sigma_i = \pm 1, \quad (1)$$

on a lattice with  $N$  sites. Suppose each site has  $2z$  nearest neighbors, so that the total number of nearest-neighbor bonds is  $Nz$ .

1. Let  $p(\sigma)$  be any trial probability distribution over spin configurations  $\sigma = \{\sigma_i\}$ . Define

$$F_{\text{var}}[p] = \sum_{\sigma} p(\sigma) H(\sigma) + T \sum_{\sigma} p(\sigma) \log p(\sigma). \quad (2)$$

Show that the exact Helmholtz free energy

$$F = -T \log Z, \quad Z = \sum_{\sigma} e^{-\beta H(\sigma)}, \quad (3)$$

satisfies

$$F \leq F_{\text{var}}[p]. \quad (4)$$

When does equality hold?

Hint: define the exact canonical distribution

$$p_{\text{eq}}(\sigma) = \frac{1}{Z} e^{-\beta H(\sigma)}. \quad (5)$$

Then compute the relative entropy

$$D(p||p_{\text{eq}}) = \sum_{\sigma} p(\sigma) \log \frac{p(\sigma)}{p_{\text{eq}}(\sigma)} \geq 0. \quad (6)$$

2. Now restrict to factorized trial distributions,

$$p(\sigma) = \prod_{i=1}^N p_i(\sigma_i), \quad (7)$$

with translational symmetry

$$p_i(+1) = \frac{1+m}{2}, \quad p_i(-1) = \frac{1-m}{2}. \quad (8)$$

Show that

$$\langle \sigma_i \rangle = m, \quad \langle \sigma_i \sigma_j \rangle = m^2 \quad (9)$$

for  $i \neq j$ , and hence

$$\frac{1}{N} \langle H \rangle = -Jzm^2 - hm. \quad (10)$$

3. Compute the entropy contribution

$$T \sum_{\sigma} p(\sigma) \log p(\sigma), \quad (11)$$

and show that the variational free energy per site is

$$f_{\text{var}}(m; T, h) = -Jzm^2 - hm + T \left[ \frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right]. \quad (12)$$

4. Minimize  $f_{\text{var}}(m; T, h)$  with respect to  $m$ . Show that the stationarity condition is

$$m = \tanh \left( \frac{2Jzm + h}{T} \right). \quad (13)$$

Conclude that the mean-field critical temperature is

$$T_c = 2Jz. \quad (14)$$

5. Expand  $f_{\text{var}}(m; T, h)$  for small  $m$ , up to order  $m^4$ , and show that

$$f_{\text{var}}(m; T, h) = \text{const.} + \frac{1}{2}(T - 2Jz)m^2 + \frac{T}{12}m^4 - hm + O(m^6). \quad (15)$$

Explain how this reproduces the Landau form of the free energy.

6. Using this Landau expansion, show that for  $h = 0$  and  $T < T_c$ ,

$$m_*^2 = \frac{3(T_c - T)}{T} \quad (16)$$

near  $T_c$ , and hence

$$m_* \sim (T_c - T)^{1/2}. \quad (17)$$

Also show that for  $T > T_c$ ,

$$\chi_T = \left( \frac{\partial m}{\partial h} \right)_{h=0} = \frac{1}{T - T_c}. \quad (18)$$

Thus the variational mean-field approximation gives the same critical behavior as Landau theory.

### Solution

1. Define the exact canonical probability distribution:

$$p_{\text{eq}}(\sigma) = \frac{1}{Z} e^{-\beta H(\sigma)}$$

Taking the logarithm gives  $\log p_{\text{eq}}(\sigma) = -\log Z - \beta H(\sigma) = \beta F - \beta H(\sigma)$ , since

$F = -T \log Z$ . We evaluate the relative entropy (Kullback-Leibler divergence), which is strictly non-negative:

$$\begin{aligned}
 0 \leq D(p||p_{\text{eq}}) &= \sum_{\sigma} p(\sigma) \log \frac{p(\sigma)}{p_{\text{eq}}(\sigma)} \\
 &= \sum_{\sigma} p(\sigma) \log p(\sigma) - \sum_{\sigma} p(\sigma) \log p_{\text{eq}}(\sigma) \\
 &= \sum_{\sigma} p(\sigma) \log p(\sigma) - \sum_{\sigma} p(\sigma) [\beta F - \beta H(\sigma)] \\
 &= \sum_{\sigma} p(\sigma) \log p(\sigma) - \beta F \sum_{\sigma} p(\sigma) + \beta \sum_{\sigma} p(\sigma) H(\sigma)
 \end{aligned}$$

Since  $\sum_{\sigma} p(\sigma) = 1$ , we can multiply by  $T (= 1/\beta)$  and rearrange:

$$\begin{aligned}
 0 &\leq T \sum_{\sigma} p(\sigma) \log p(\sigma) - F + \sum_{\sigma} p(\sigma) H(\sigma) \\
 F &\leq \sum_{\sigma} p(\sigma) H(\sigma) + T \sum_{\sigma} p(\sigma) \log p(\sigma) = F_{\text{var}}[p]
 \end{aligned}$$

Equality holds if and only if the relative entropy is zero, which happens when the trial distribution is exactly the canonical distribution:  $p(\sigma) = p_{\text{eq}}(\sigma)$  for all  $\sigma$ .

2. For a factorized distribution  $p(\sigma) = \prod_k p_k(\sigma_k)$  with  $p_i(\pm 1) = \frac{1 \pm m}{2}$ :

$$\langle \sigma_i \rangle = (+1) \frac{1+m}{2} + (-1) \frac{1-m}{2} = \frac{1+m-1+m}{2} = m$$

Because the distribution factors, variables at different sites are completely independent. Thus, for  $i \neq j$ :

$$\langle \sigma_i \sigma_j \rangle = \langle \sigma_i \rangle \langle \sigma_j \rangle = m \cdot m = m^2$$

The expectation value of the Hamiltonian is:

$$\langle H \rangle = -J \sum_{\langle ij \rangle} \langle \sigma_i \sigma_j \rangle - h \sum_i \langle \sigma_i \rangle$$

There are  $Nz/2$  pairs of nearest neighbors in the lattice (since each of  $N$  sites has  $2z$  neighbors, and each bond connects 2 sites).

$$\langle H \rangle = -J \left( \frac{Nz}{2} \right) m^2 - hNm$$

Wait, the problem states the total number of bonds is  $Nz$ . Let's assume the sum  $\sum_{\langle ij \rangle}$  runs over all directed bonds  $i \rightarrow j$ , which would be  $N \cdot (2z)$  terms? No, the notation  $\sum_{\langle ij \rangle}$  usually means sum over pairs. If the lattice has coordination number  $2z$ , the number of pairs is  $Nz$ . Let's follow the problem's hint that there are  $Nz$  bonds. Then:

$$\langle H \rangle = -J(Nz)m^2 - hNm \implies \frac{1}{N} \langle H \rangle = -Jzm^2 - hm$$

3. For a factorized distribution, the entropy separates into a sum over independent sites:

$$\sum_{\sigma} p(\sigma) \log p(\sigma) = \sum_i \sum_{\sigma_i} p_i(\sigma_i) \log p_i(\sigma_i)$$

For a single site  $i$ :

$$\sum_{\sigma_i} p_i(\sigma_i) \log p_i(\sigma_i) = p_i(+1) \log p_i(+1) + p_i(-1) \log p_i(-1) = \frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2}$$

Since all  $N$  sites are identical:

$$\frac{T}{N} \sum_{\sigma} p(\sigma) \log p(\sigma) = T \left[ \frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right]$$

Adding the energy per site derived in part 2, the variational free energy per site is:

$$f_{\text{var}}(m; T, h) = \frac{F_{\text{var}}}{N} = -Jzm^2 - hm + T \left[ \frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right]$$

4. We minimize  $f_{\text{var}}(m; T, h)$  with respect to  $m$ :

$$\begin{aligned} \frac{\partial f_{\text{var}}}{\partial m} &= -2Jzm - h + \frac{T}{2} \left[ \log \frac{1+m}{2} + 1 - \log \frac{1-m}{2} - 1 \right] = 0 \\ -2Jzm - h + \frac{T}{2} \log \left( \frac{1+m}{1-m} \right) &= 0 \end{aligned}$$

Rearranging gives:

$$\frac{1+m}{1-m} = \exp \left( \frac{2(2Jzm + h)}{T} \right)$$

Solving for  $m$  using the identity  $\frac{e^{2x}-1}{e^{2x}+1} = \tanh x$  directly yields:

$$m = \tanh \left( \frac{2Jzm + h}{T} \right)$$

At  $h = 0$ , a non-zero solution  $m \neq 0$  emerges when the slope of  $\tanh(2Jzm/T)$  at origin exceeds 1. Since  $\frac{d}{dm} \tanh(cx)|_{m=0} = c$ , the critical condition is  $2Jz/T_c = 1$ , which gives the mean-field critical temperature:

$$T_c = 2Jz$$

5. To expand  $f_{\text{var}}(m)$  for small  $m$ , we use the Taylor expansion of  $\log(1 \pm m)$ :

$$\log(1 \pm m) = \pm m - \frac{m^2}{2} \pm \frac{m^3}{3} - \frac{m^4}{4} + O(m^5)$$

The entropy term becomes:

$$\begin{aligned} \frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} &= -\log 2 + \frac{1+m}{2} \log(1+m) + \frac{1-m}{2} \log(1-m) \\ &= -\log 2 + \frac{1}{2} [\log(1+m) + \log(1-m)] + \frac{m}{2} [\log(1+m) - \log(1-m)] \\ &= -\log 2 + \frac{1}{2} \left[ -m^2 - \frac{m^4}{2} \right] + \frac{m}{2} \left[ 2m + \frac{2m^3}{3} \right] \\ &= -\log 2 - \frac{m^2}{2} - \frac{m^4}{4} + m^2 + \frac{m^4}{3} = -\log 2 + \frac{m^2}{2} + \frac{m^4}{12} + O(m^6) \end{aligned}$$

Putting this back into  $f_{\text{var}}$ :

$$\begin{aligned} f_{\text{var}}(m; T, h) &= -T \log 2 - Jzm^2 - hm + T \left( \frac{m^2}{2} + \frac{m^4}{12} \right) \\ &= -T \log 2 + \frac{1}{2}(T - 2Jz)m^2 + \frac{T}{12}m^4 - hm + O(m^6) \end{aligned}$$

This precisely reproduces the Landau form of the free energy  $\phi = \phi_0 + \frac{at}{2}m^2 + \frac{u}{4}m^4 - hm$ . By identifying  $t = T - T_c$  (where  $T_c = 2Jz$ ), we see the  $m^2$  coefficient changes sign exactly at the critical point  $T = T_c$ , driving the continuous phase transition.

6. Set  $h = 0$ . The stationary points of the expanded free energy are given by  $\partial f_{\text{var}}/\partial m = 0$ :

$$(T - T_c)m + \frac{T}{3}m^3 = 0$$

For  $T < T_c$ , the non-zero minimum is found by dividing by  $m$ :

$$\frac{T}{3}m^2 = T_c - T \implies m_*^2 = \frac{3(T_c - T)}{T}$$

Since  $T \approx T_c$  near the transition,  $m_* \sim (T_c - T)^{1/2}$ , yielding the mean-field exponent  $\beta = 1/2$ .

For  $T > T_c$  with  $h \neq 0$ , the equilibrium condition is:

$$(T - T_c)m + \frac{T}{3}m^3 - h = 0$$

Differentiate implicitly with respect to  $h$  evaluated at  $h = 0$  (where  $m = 0$  for  $T > T_c$ ):

$$(T - T_c)\frac{\partial m}{\partial h} + Tm^2\frac{\partial m}{\partial h} - 1 = 0 \implies \chi_T = \left. \frac{\partial m}{\partial h} \right|_{h=0, m=0} = \frac{1}{T - T_c}$$

This yields the critical exponent  $\gamma = 1$ . The variational mean-field approximation produces exact Landau exponents.

### Mean-field theory near a tricritical point

Consider the Landau free energy

$$\phi(m; t, u, h) = \phi_0 + \frac{at}{2}m^2 + \frac{u}{4}m^4 + \frac{v}{6}m^6 - hm, \quad a > 0, \quad v > 0. \quad (19)$$

Here  $t$  is the temperature-like variable,  $h$  is the field conjugate to the order parameter  $m$ , and  $u$  is an additional control parameter. The case  $u > 0$  gives the ordinary mean-field critical point. The point

$$t = 0, \quad u = 0, \quad h = 0$$

is called a tricritical point.

1. Explain why the  $m^6$  term is needed when  $u < 0$ . What would go wrong if one kept only terms up to  $m^4$ ?
2. Set  $h = 0$ . Show that the stationary points satisfy

$$m(at + um^2 + vm^4) = 0.$$

Writing  $q = m^2$ , show that the nonzero stationary points satisfy

$$vq^2 + uq + at = 0.$$

3. Assume  $u > 0$ . Show that the transition at  $t = 0$  is continuous. Near  $t = 0^-$ , show that

$$m^2 = -\frac{at}{u} + O(t^2).$$

Conclude that the  $m^6$  term changes only subleading corrections and does not affect the ordinary mean-field exponent  $\beta = 1/2$ .

4. Assume  $u < 0$ . At coexistence, the minimum at  $m = 0$  and the nonzero minima have equal free energy. Use the two conditions

$$at + um^2 + vm^4 = 0,$$

and

$$\frac{at}{2}m^2 + \frac{u}{4}m^4 + \frac{v}{6}m^6 = 0$$

to show that

$$m^2 = -\frac{3u}{4v}, \quad t_{\text{coex}}(u) = \frac{3u^2}{16av}.$$

Sketch the phase diagram in the  $(u, t)$  plane. Label the continuous transition line

$$t = 0, \quad u > 0,$$

the first-order transition line

$$t = \frac{3u^2}{16av}, \quad u < 0,$$

and the tricritical point

$$(t, u) = (0, 0).$$

5. Now set  $u = 0$ . At  $h = 0$ , show that for  $t < 0$ ,

$$m^4 = -\frac{at}{v}.$$

Hence determine the tricritical exponent  $\beta_{\text{tri}}$ , defined by

$$m(t, 0) \sim (-t)^{\beta_{\text{tri}}}.$$

6. At the tricritical point  $t = 0$ ,  $u = 0$ , show that the equation of state is

$$h = vm^5.$$

Determine the tricritical exponent  $\delta_{\text{tri}}$ , defined by

$$m(0, h) \sim h^{1/\delta_{\text{tri}}}.$$

7. Still at  $u = 0$ , compute the susceptibility exponent. Starting from

$$h = atm + vm^5,$$

show that

$$\chi_T^{-1} = \left( \frac{\partial h}{\partial m} \right)_t = at + 5vm^4.$$

Use this result above and below  $T_c$  to show that

$$\gamma_{\text{tri}} = 1.$$

8. At  $u = 0$  and  $h = 0$ , substitute the equilibrium value of  $m$  into the Landau free energy and show that

$$f_s(t, 0) \sim -|t|^{3/2} \quad (t < 0).$$

Determine the tricritical exponent  $\alpha_{\text{tri}}$ , defined by

$$f_s(t, 0) \sim |t|^{2-\alpha_{\text{tri}}}.$$

9. Finally, show directly that at  $u = 0$ , the singular free energy has the scaling form

$$f_s(t, h) = |t|^{3/2} \Psi_{\pm} \left( \frac{h}{|t|^{5/4}} \right).$$

Identify the gap exponent  $\Delta_{\text{tri}}$ , and verify that

$$\Delta_{\text{tri}} = \beta_{\text{tri}} \delta_{\text{tri}}.$$

### Solution

1. If  $u < 0$  and we keep terms only up to  $m^4$ , the free energy  $\phi(m)$  would diverge to  $-\infty$  as  $m \rightarrow \pm\infty$ . The system would lack a stable thermodynamic global minimum. Adding a term with a positive coefficient for the next highest even power, specifically  $vm^6/6$  with  $v > 0$ , ensures the free energy goes to  $+\infty$  for large  $|m|$ , stabilizing the free energy and allowing thermodynamic equilibrium.
2. Setting  $h = 0$ , the stationary points of  $\phi(m; t, u, 0)$  are:

$$\frac{\partial\phi}{\partial m} = atm + um^3 + vm^5 = m(at + um^2 + vm^4) = 0$$

By writing  $q = m^2$ , the term in the parenthesis gives the condition for non-zero stationary points ( $m \neq 0$ ):

$$vq^2 + uq + at = 0$$

3. If  $u > 0$ , at  $t = 0$ , the equation  $vq^2 + uq = 0$  has roots  $q = 0$  and  $q = -u/v < 0$  (unphysical since  $q = m^2 \geq 0$ ). Thus  $m = 0$  is the unique physical minimum, and the transition is continuous. Near  $t = 0^-$ , we have small  $q > 0$ . We solve the quadratic equation for  $q$ :

$$q = \frac{-u + \sqrt{u^2 - 4avt}}{2v} = \frac{u}{2v} \left( -1 + \left( 1 - \frac{4avt}{u^2} \right)^{1/2} \right)$$

Taylor expanding the square root  $\sqrt{1-x} \approx 1 - x/2 + O(x^2)$ :

$$q \approx \frac{u}{2v} \left( -1 + 1 - \frac{2avt}{u^2} \right) = -\frac{at}{u}$$

Hence  $m^2 = -at/u + O(t^2)$ , meaning  $m \sim (-t)^{1/2}$ . The exponent remains  $\beta = 1/2$ . The  $m^6$  term ( $v > 0$ ) only modifies higher-order terms  $O(t^2)$  and preserves standard mean-field behavior.

4. Assume  $u < 0$ . At a first-order phase coexistence, the non-zero minimum must have the exact same free energy as the minimum at  $m = 0$  (where  $\phi = \phi_0$ ). So  $\phi(m) - \phi_0 = 0$ :

$$\frac{at}{2}m^2 + \frac{u}{4}m^4 + \frac{v}{6}m^6 = 0$$

And the point must be a minimum, satisfying:

$$at + um^2 + vm^4 = 0$$

We eliminate  $at$  by substituting  $at = -um^2 - vm^4$  into the first equation (after dividing by  $m^2/2 \neq 0$ ):

$$(-um^2 - vm^4) + \frac{u}{2}m^2 + \frac{v}{3}m^4 = 0$$

$$-\frac{u}{2}m^2 - \frac{2v}{3}m^4 = 0 \implies m^2 = -\frac{3u}{4v}$$

To find the coexistence temperature, substitute  $m^2$  back into the stationarity equation:

$$at = -u \left( -\frac{3u}{4v} \right) - v \left( -\frac{3u}{4v} \right)^2 = \frac{3u^2}{4v} - \frac{9u^2}{16v} = \frac{3u^2}{16v}$$

Thus,  $t_{\text{coex}}(u) = \frac{3u^2}{16av}$ . **Phase diagram sketch logic:** The  $u$ - $t$  plane is divided. The line  $t = 0$  for  $u > 0$  is a continuous transition line. The curve  $t = 3u^2/(16av)$  for  $u < 0$  is a first-order transition line. They meet perfectly at the origin  $(0, 0)$ , which acts as the tricritical point separating the first-order and second-order regimes.

5. Now set  $u = 0$  (sitting exactly at the tricritical value of  $u$ ). For  $h = 0$  and  $t < 0$ , the stationary equation  $at + um^2 + vm^4 = 0$  strictly simplifies to:

$$at + vm^4 = 0 \implies m^4 = -\frac{at}{v}$$

Thus,  $m = \left(\frac{a}{v}\right)^{1/4} (-t)^{1/4} \sim (-t)^{1/4}$ . The tricritical exponent is  $\beta_{\text{tri}} = 1/4$ .

6. At the tricritical point  $t = 0, u = 0$ , the state equation  $\partial\phi/\partial m = 0$  with an external field  $h$  becomes:

$$vm^5 - h = 0 \implies h = vm^5$$

Solving for  $m$  yields  $m \sim h^{1/5}$ . By the definition  $m \sim h^{1/\delta_{\text{tri}}}$ , we get  $\delta_{\text{tri}} = 5$ .

7. Still at  $u = 0$ , the equation of state is  $h = atm + vm^5$ . The inverse isothermal susceptibility is:

$$\chi_T^{-1} = \left( \frac{\partial h}{\partial m} \right)_t = at + 5vm^4$$

For  $T > T_c$  ( $t > 0$ ),  $m = 0$ , so  $\chi_T^{-1} = at \implies \chi_T \sim t^{-1}$ , giving  $\gamma_{\text{tri}}^+ = 1$ . For  $T < T_c$  ( $t < 0$ ),  $m^4 = -at/v$ . Substituting this in:

$$\chi_T^{-1} = at + 5v \left( -\frac{at}{v} \right) = at - 5at = -4at$$

Since  $t < 0$ ,  $\chi_T \sim (-t)^{-1}$ , giving  $\gamma_{\text{tri}}^- = 1$ . Hence  $\gamma_{\text{tri}} = 1$ .

8. At  $u = 0, h = 0$  and  $t < 0$ , we have  $m^2 = (-at/v)^{1/2}$ . Substitute into the free energy to find the singular part:

$$\begin{aligned} f_s(t, 0) &= \frac{at}{2}m^2 + \frac{v}{6}m^6 = \frac{at}{2} \left( -\frac{at}{v} \right)^{1/2} + \frac{v}{6} \left( -\frac{at}{v} \right)^{3/2} \\ &= -\frac{1}{2} \frac{a^{3/2}}{v^{1/2}} (-t)^{3/2} - \frac{1}{6} \frac{a^{3/2}}{v^{1/2}} (-t)^{3/2} = -\frac{2}{3} \frac{a^{3/2}}{v^{1/2}} (-t)^{3/2} \end{aligned}$$

Thus  $f_s(t, 0) \sim -|t|^{3/2}$ . By  $f_s \sim |t|^{2-\alpha_{\text{tri}}}$ , we get  $2 - \alpha_{\text{tri}} = 3/2 \implies \alpha_{\text{tri}} = 1/2$ .

9. The equation of state at  $u = 0$  is  $h = atm + vm^5$ . Let's scale  $m = |t|^{1/4}y$ .

$$h = at(|t|^{1/4}y) + v(|t|^{1/4}y)^5 = \pm a|t|^{5/4}y + v|t|^{5/4}y^5 = |t|^{5/4}(\pm ay + vy^5)$$

This implies  $y$  is a function of the scaled variable  $x = h/|t|^{5/4}$ . The singular free energy is  $f_s = \frac{at}{2}m^2 + \frac{v}{6}m^6 - hm$ . Substitute  $m = |t|^{1/4}y$ :

$$f_s = \frac{at}{2}|t|^{1/2}y^2 + \frac{v}{6}|t|^{3/2}y^6 - h|t|^{1/4}y = |t|^{3/2} \left[ \pm \frac{a}{2}y^2 + \frac{v}{6}y^6 - \frac{h}{|t|^{5/4}}y \right]$$

Because  $y$  depends strictly on  $h/|t|^{5/4}$ , we can write  $f_s = |t|^{3/2}\Psi_{\pm}(h/|t|^{5/4})$ . The gap exponent  $\Delta_{\text{tri}}$  is the power in the scaling argument  $h/|t|^{\Delta_{\text{tri}}}$ , so  $\Delta_{\text{tri}} = 5/4$ . Checking the relation  $\beta_{\text{tri}}\delta_{\text{tri}}$ :

$$\beta_{\text{tri}}\delta_{\text{tri}} = \left(\frac{1}{4}\right) \times 5 = \frac{5}{4} = \Delta_{\text{tri}}$$

This perfectly verifies the scaling identity.

### Critical exponents from a linearized RG transformation

Consider a simple real-space RG transformation for two variables  $t$  and  $h$ :

$$t' = 2t - t^2, \quad h' = 4h + th.$$

Here  $t$  is a temperature-like perturbation and  $h$  is a field-like perturbation. The RG scale factor is  $b = 2$ . This map is not meant to be an exact RG transformation of the Ising model; it is a simple model for the linearization procedure near a critical fixed point.

1. Find the fixed point of the RG transformation near the origin.
2. Linearize the transformation near this fixed point. Find the eigenvalues of the linearized RG transformation.
3. Use the eigenvalues to determine the RG scaling dimensions  $y_t$  and  $y_h$ , defined by

$$t' = b^{y_t} t, \quad h' = b^{y_h} h$$

to linear order.

4. Assume the spatial dimension is  $d = 3$ . Using the scaling relations

$$\begin{aligned} \nu &= \frac{1}{y_t}, & \alpha &= 2 - \frac{d}{y_t}, \\ \beta &= \frac{d - y_h}{y_t}, & \gamma &= \frac{2y_h - d}{y_t}, \\ \delta &= \frac{y_h}{d - y_h}, & \eta &= d + 2 - 2y_h, \end{aligned}$$

determine all six critical exponents

$$\alpha, \quad \beta, \quad \gamma, \quad \delta, \quad \nu, \quad \eta.$$

### Solution

1. A fixed point  $(t^*, h^*)$  of the RG transformation occurs when  $t' = t$  and  $h' = h$ .

$$t^* = 2t^* - (t^*)^2 \implies (t^*)^2 - t^* = 0 \implies t^* = 0 \text{ or } 1$$

Since we are looking for the fixed point \*near the origin\*, we choose  $t^* = 0$ .

$$h^* = 4h^* + t^*h^* \implies h^* = 4h^* \implies 3h^* = 0 \implies h^* = 0$$

Thus, the relevant fixed point is exactly at the origin  $(t^*, h^*) = (0, 0)$ .

2. To linearize the transformation near  $(0, 0)$ , we construct the Jacobian matrix  $M$  evaluated at the fixed point:

$$M = \begin{pmatrix} \frac{\partial t'}{\partial t} & \frac{\partial t'}{\partial h} \\ \frac{\partial h'}{\partial t} & \frac{\partial h'}{\partial h} \end{pmatrix}_{(0,0)}$$

Calculating the partial derivatives:

$$\begin{aligned}\frac{\partial t'}{\partial t} &= 2 - 2t \xrightarrow{t=0} 2, & \frac{\partial t'}{\partial h} &= 0 \\ \frac{\partial h'}{\partial t} &= h \xrightarrow{h=0} 0, & \frac{\partial h'}{\partial h} &= 4 + t \xrightarrow{t=0} 4\end{aligned}$$

The linearized RG transformation matrix is strictly diagonal:

$$M = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

The eigenvalues of this diagonal matrix are obviously  $\Lambda_t = 2$  and  $\Lambda_h = 4$ .

3. The eigenvalues are connected to the RG scaling dimensions by  $\Lambda = b^y$ , where  $b = 2$  is the spatial rescaling factor. For the temperature perturbation:

$$\Lambda_t = 2 \implies 2^{y_t} = 2 \implies y_t = 1$$

For the field perturbation:

$$\Lambda_h = 4 \implies 2^{y_h} = 4 \implies y_h = 2$$

4. We use the established scaling dimensions  $y_t = 1$  and  $y_h = 2$  in spatial dimension  $d = 3$  to calculate the six critical exponents:

- **Correlation length exponent  $\nu$ :**

$$\nu = \frac{1}{y_t} = \frac{1}{1} = 1$$

- **Specific heat exponent  $\alpha$ :**

$$\alpha = 2 - \frac{d}{y_t} = 2 - \frac{3}{1} = -1$$

- **Order parameter exponent  $\beta$ :**

$$\beta = \frac{d - y_h}{y_t} = \frac{3 - 2}{1} = 1$$

- **Susceptibility exponent  $\gamma$ :**

$$\gamma = \frac{2y_h - d}{y_t} = \frac{2(2) - 3}{1} = 1$$

- **Critical isotherm exponent  $\delta$ :**

$$\delta = \frac{y_h}{d - y_h} = \frac{2}{3 - 2} = 2$$

- **Anomalous dimension  $\eta$ :**

$$\eta = d + 2 - 2y_h = 3 + 2 - 2(2) = 5 - 4 = 1$$

These constitute the complete set of critical exponents derived from the given RG relations.